

ZINC ABSORPTION AND ADHERENCE TO THE EAT-LANCET PLANETARY HEALTH DIET

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INTRODUCTION

Emphasis is increasingly placed on incorporating environmental sustainability into dietary guidelines. The 2025 EAT-Lancet Planetary Health Diet (PHD) contains a set of flexible dietary recommendations, described as nutritionally adequate for most nutrients.⁽¹⁾ However, we previously found that one-third of Australian adults had inadequate zinc intakes, and that higher PHD adherence did not protect against this.⁽²⁾ Because the PHD is characterised by higher plant-based food intakes, it may also be associated with higher phytate intakes, which can reduce zinc absorption.⁽³⁾ Few studies have addressed zinc absorption when assessing PHD adherence;⁽⁴⁾ in Australia, this is yet to be examined. We aimed to 1) assess the prevalence of inadequate zinc intake, 2) estimate the absorption of zinc and 3) examine associations between PHD adherence and total absorbable zinc (TAZ).

METHODS

Dietary data from the 2011-2013 National Nutrition and Physical Activity Survey were analysed from n = 5,655 adults aged ≥ 18 years. Individuals were excluded if they consumed dietary supplements, were pregnant or lactating, or misreported energy intake.

Dietary intakes were assessed using two 24-hour dietary recalls. Usual food and nutrient intakes were modelled using the Multiple Source Method. Inadequate zinc intake was assessed using the Estimated Average Requirement cut-point method. TAZ (mg/day) was estimated using a multivariate saturation algorithm;⁽⁵⁾ TAZ adequacy was determined using the Institute of Medicine’s sex-specific physiological requirements for TAZ.

The Healthy Reference Diet Score (HRDS) was used to assess PHD adherence (0 points represent the lowest score, and 140, the highest) according to the usual grams of foods consumed per day.⁽²⁾ Logistic and linear regression models assessed associations between HRDS and TAZ; restricted cubic splines tested non-linearity.

All models were adjusted for age, sex and energy intake and jackknife survey weights were applied throughout.

RESULTS AND DISCUSSION

The mean (SE) HRDS was 50 (0.3) and ranged from 6 to 104 points. Mean (SE) zinc, protein and calcium intake was 10.7 (0.1) mg/day, 89.8 (0.4) g/day and 776.3 (5.3) mg/day, respectively. Median (25th, 75th percentile) phytate intake was 519.7 (370.9, 727.2) mg/day among those in the lowest HRDS quartile, compared with 755.8 (547, 1004.3) mg/day in the highest. The prevalence of inadequate zinc intake was 35% overall, ranging from 42% in the lowest HRDS quartile to 31% in the highest. Each 10-point increase in HRDS (Table 1) was associated with less mg/day of TAZ ($\beta = -0.074$; 95% CI: $-0.089, -0.060$; $p < 0.001$). Similarly, HRDS was associated with greater odds of inadequate TAZ (OR = 1.36; 95% CI: 1.274, 1.450; $p < 0.001$) per 10-point increase.

CONCLUSIONS

Zinc intakes require further attention in Australia; those with higher levels of PHD adherence may be at greater risk of absorbing inadequate zinc. Communicating zinc requirements for the PHD may require more nuanced consideration, similar to vegetarian diets.

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Table 1. Survey-Weighted Logistic and Linear Regression Models of 2013 NNPAS Assessing Associations between Total Absorbable Zinc and Adherence to the 2025 EAT-Lancet Planetary Health Diet Per 10-point Increase in Adherence Score.

Logistic Regression	Odds Ratio	95% CI	p-Value
TAZ (inadequate; reference) ¹	1.360	1.274, 1.450	$p < 0.001$
Linear Regression	β Coefficient	95% CI	p-Value
TAZ (mg/day)	-0.074	-0.089, -0.060	$p < 0.001$

NNPAS, National Nutrition and Physical Activity Survey. TAZ, Total Absorbable Zinc. ¹ Inadequacy of TAZ was defined as intakes below the Institute of Medicine’s physiological requirements for absorbed zinc (3.3 mg/day for women and 3.8 mg/day for men). These values are the average minimum amounts required to be absorbed to replace endogenous zinc losses and excretion. (8)